Z-motion correction

validation & benchmarking

optimouse 2-photon pipeline — C57_O1M2/10022023

CaImAn 1.12.2 · Suite2p 0.14.0 · real 41-plane z-stack

1. Overview: what is validated, and how

Validation of z-motion correction on real anatomy + synthetic ground truth

The z-motion correction chain is: motion correction → z-position estimation → F_anat construction → ROI extraction → per-ROI fit-and-subtract. It removes the apparent-fluorescence changes caused by axial (z) brain motion, which defocuses cells and modulates their signal independently of neural activity.

Two data regimes are used throughout this report:

- Real data (C57_O1M2/10022023): functional TSeries + a real 41-plane z-stack (1 µm steps). This validates the end-to-end path and exposes production behaviour (both CNMF and Suite2p extraction backends run), but has no ground truth to score against.

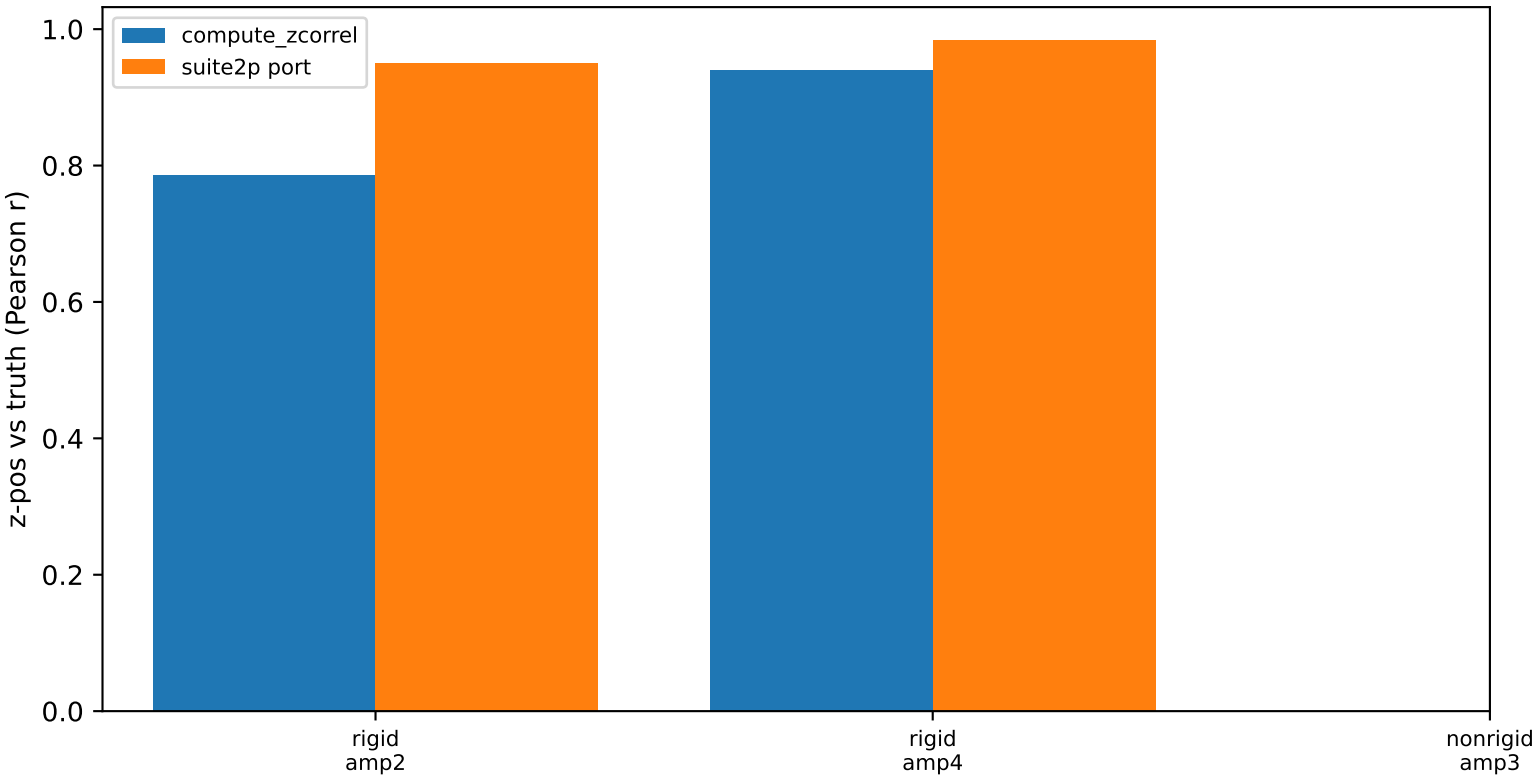
- Synthetic ground truth: a NAOMi-lite harness that takes the real z-stack as the anatomical volume, injects known spike→GCaMP activity into detected somata, and samples the volume along a KNOWN 3-D trajectory (a staircase z-path + xy jitter + mild deformation). An axial-Gaussian gain models defocus, so z-motion attenuates cells — reproducing the artifact. Because the true activity and z-position are known, this quantifies both estimation error and recovery. The z-motion is response-coupled: downward steps cluster on stimulus responses, as in the real protocol (the animal reacts, so the brain moves when cells respond) — the hardest case, since the artifact then overlaps the signal.

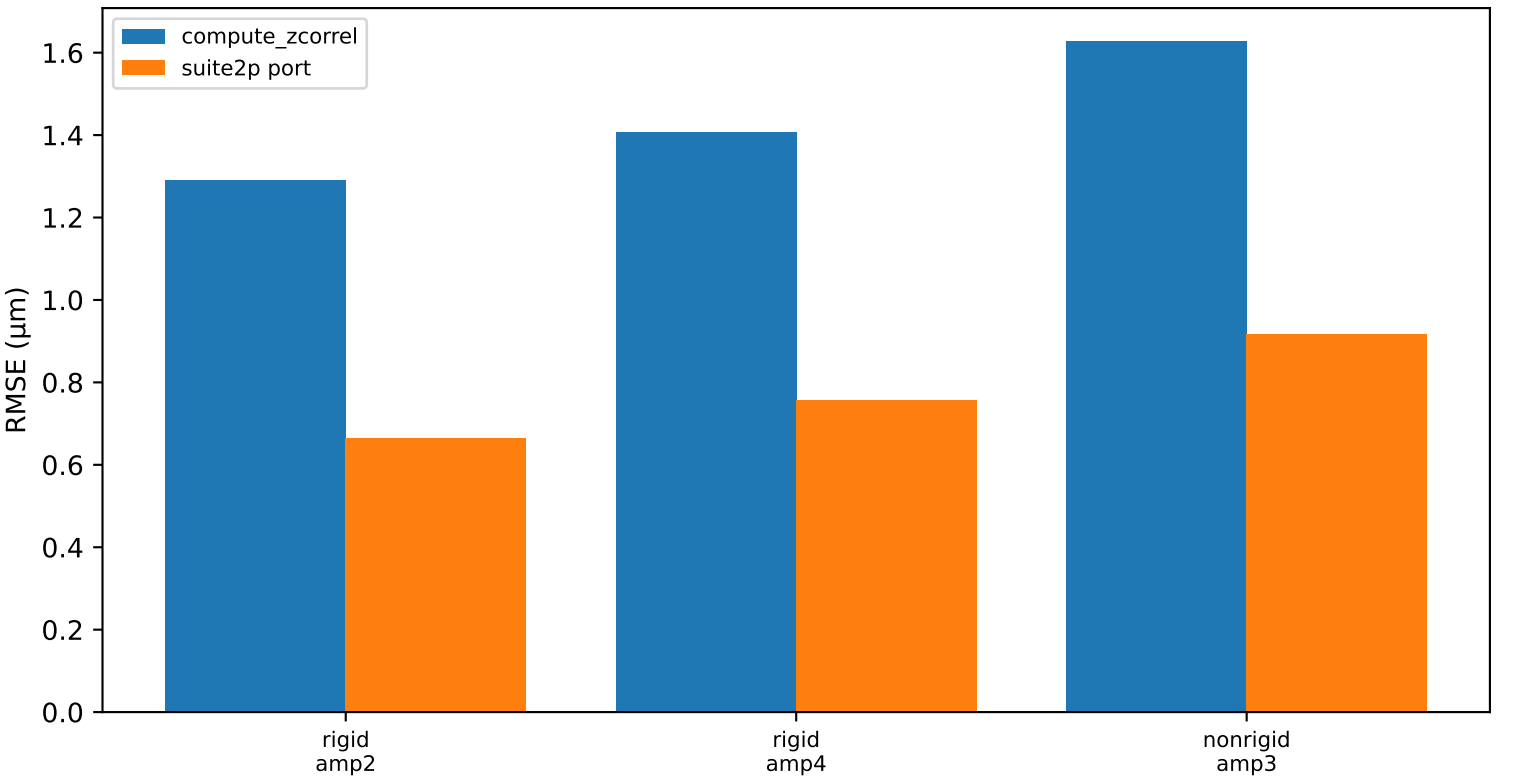
Two things are compared:

- z-position estimation (§2–3): full-frame Pearson correlation (shipped) vs Suite2p masked phase-correlation against the z-stack (sub-plane interpolation in Annex A2). Tested on synthetic ground truth (accuracy vs a known trajectory) AND on real motion-corrected recordings (stability, single + concatenated sessions).

- z-motion correction (§4–8): the F_anat fit-and-subtract applied at three spatial granularities — per-pixel, per-patch, and per-ROI. The shipped per-ROI method follows the MATLAB reference (project F_anat onto each ROI footprint, OLS-regress ΔF/F on it, subtract). Compared on synthetic ground truth (recovery of the true activity, residual z-contamination) and shown on real data.

2. z-position — synthetic method sweep (accuracy vs drift amplitude)





Suite2p is more accurate (higher r, lower RMSE) but slower — time/case: rigid amp2: 13.8s vs 82.6s rigid amp4: 2.6s vs 7.9s nonrigid amp3: 2.8s vs 8.0s

3. z-position: Pearson (shipped) vs Suite2p

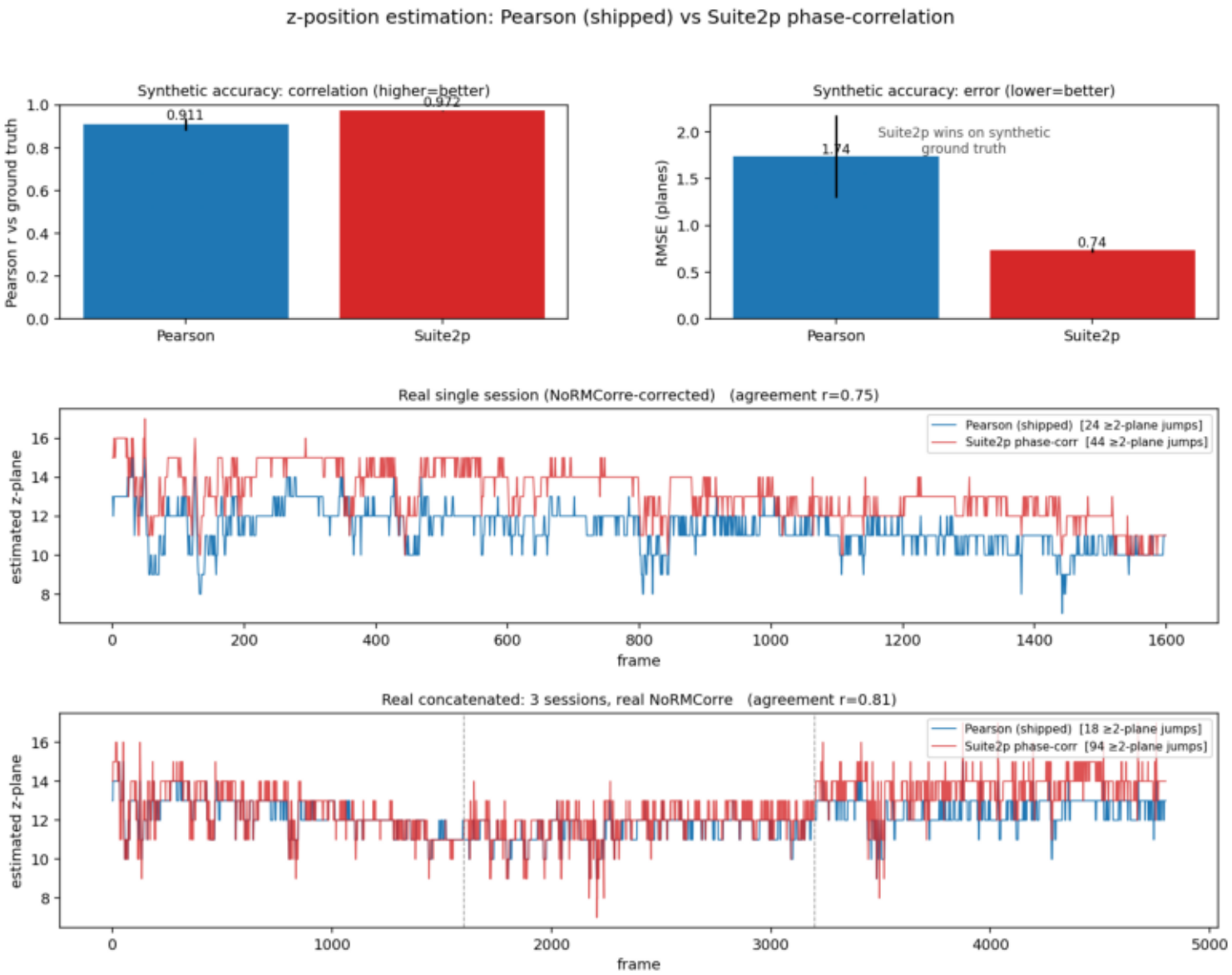
Head-to-head of the two z-position estimators: the shipped Pearson full-frame correlation (+ temporal median smoothing) vs Suite2p masked phase-correlation.

A. On SYNTHETIC ground truth Suite2p is more accurate (r 0.972 vs 0.911, RMSE 0.74 vs 1.74 planes).

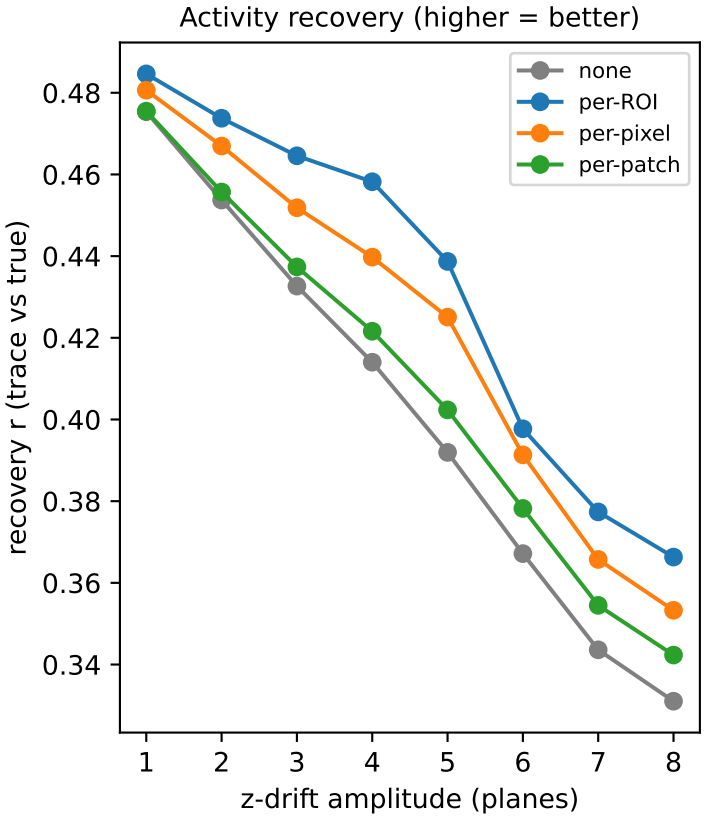
B/C. On REAL NoRMCorre output the ranking reverses: phase-correlation is much noisier (and biased toward deep, low-contrast planes in single sessions), because mcorr edge artifacts (border_nan="copy") corrupt the masked correlation. Pearson + temporal smoothing tracks a stable trajectory across sessions (dashed = session boundaries).

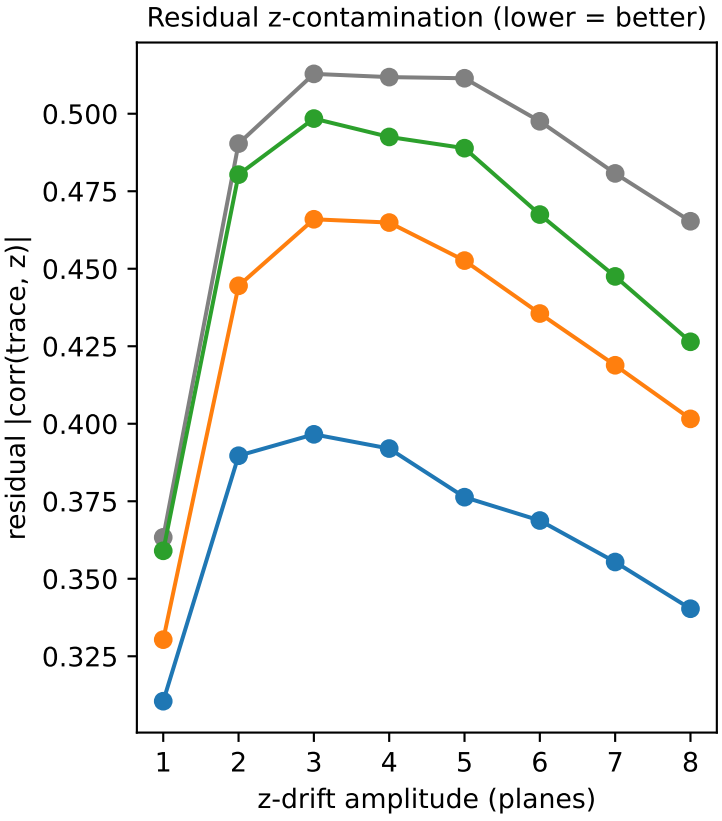
On the 3-session real-NoRMCorre concat (C), phase-correlation makes 94 spurious ≥2-plane jumps vs Pearson's 18 (5× more) while tracking the same overall drift (agreement r=0.81).

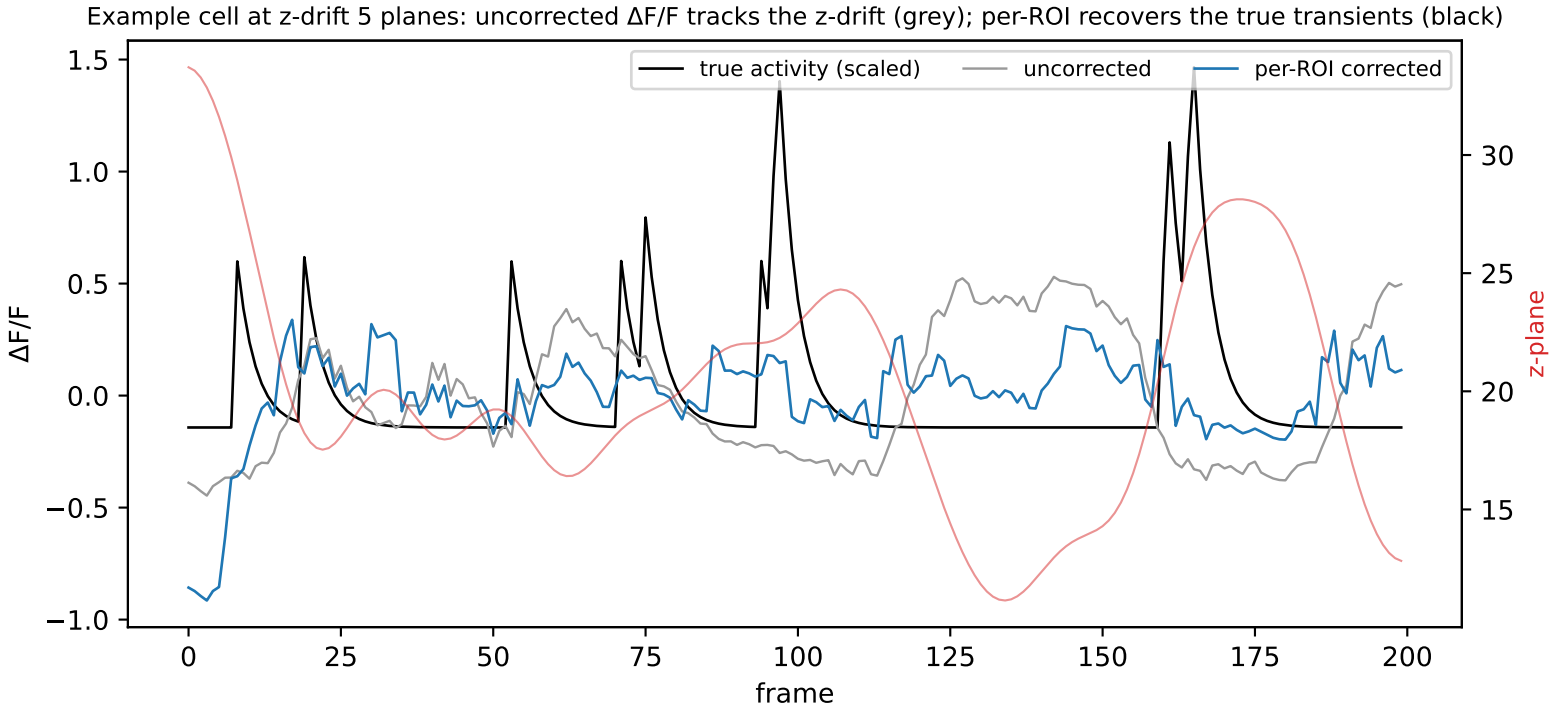
Since z-position is estimated AFTER motion correction in production, the real-data behaviour is what matters → the pipeline ships Pearson (no Suite2p dependency).



4. Correction — method comparison (synthetic ground truth)







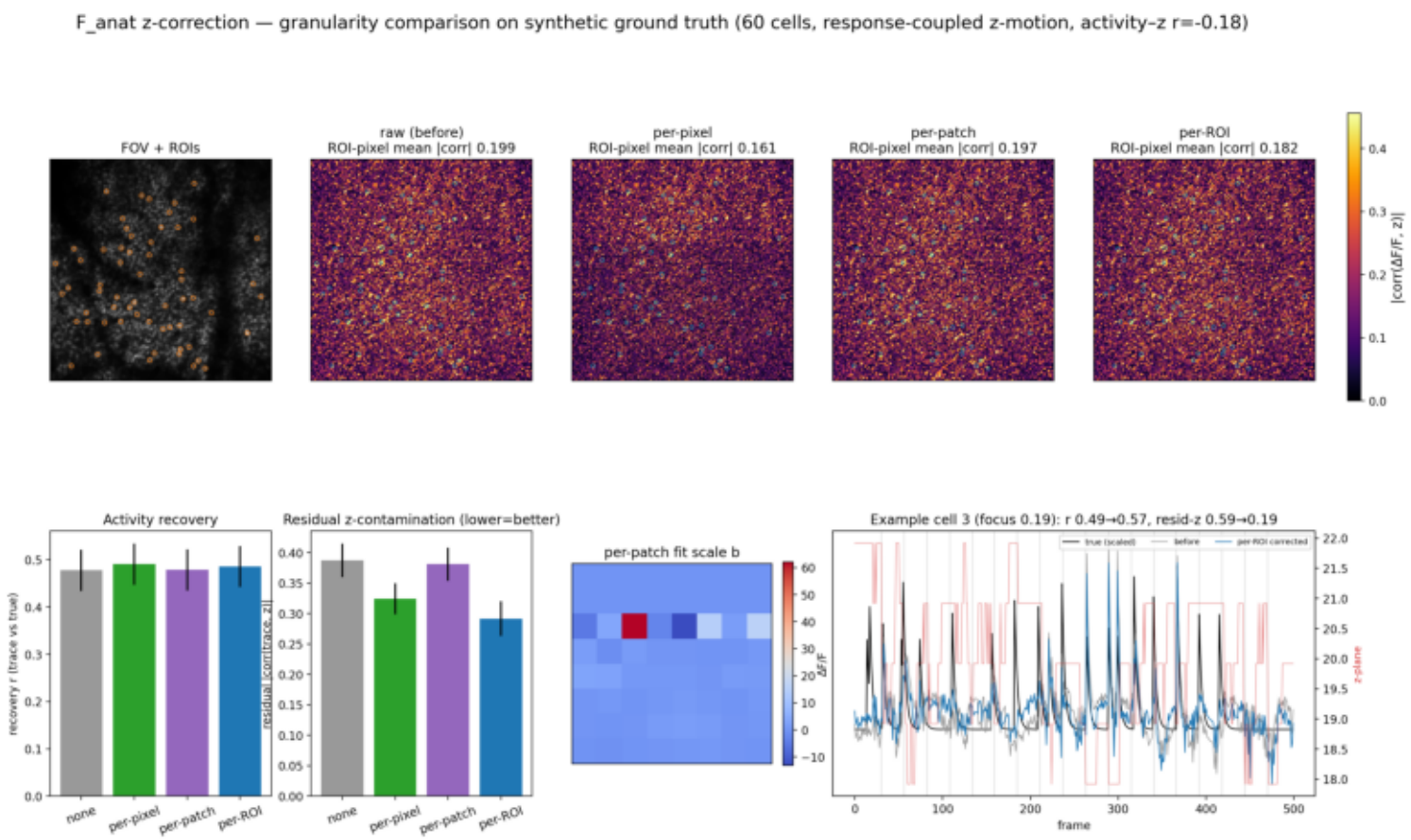
Measured on synthetic ground truth: known activity injected into the real z-stack under a rigid z-drift of each amplitude. Per-ROI removes the most z-contamination at every amplitude; recovery is ultimately capped by defocus attenuation of the true signal.

5. Correction granularity: pixels vs patches vs ROI (synthetic)

The F_anat z-correction can be applied at three spatial granularities. The top row maps per-pixel residual z-contamination |corr(ΔF/F, z)| over the FOV after each method (cyan = ROI centers); the bottom row quantifies recovery and residual z on the extracted ROI traces. All quantities are measured on synthetic ground truth.

- per-pixel — regress every pixel on F_anat (a-priori; over-corrects, mixes noise).
- per-patch — one scale per spatial patch (coarse).
- per-ROI — one OLS fit per cell footprint (the shipped method).

Residual z-contamination on traces: none 0.39 → per-pixel 0.32 → per-patch 0.38 → per-ROI 0.29 (lower is better). Per-ROI removes the most z from the extracted traces while preserving activity — per-pixel looks lowest pixel-wise but over-corrects, so the ROI-mean trace keeps more residual z.



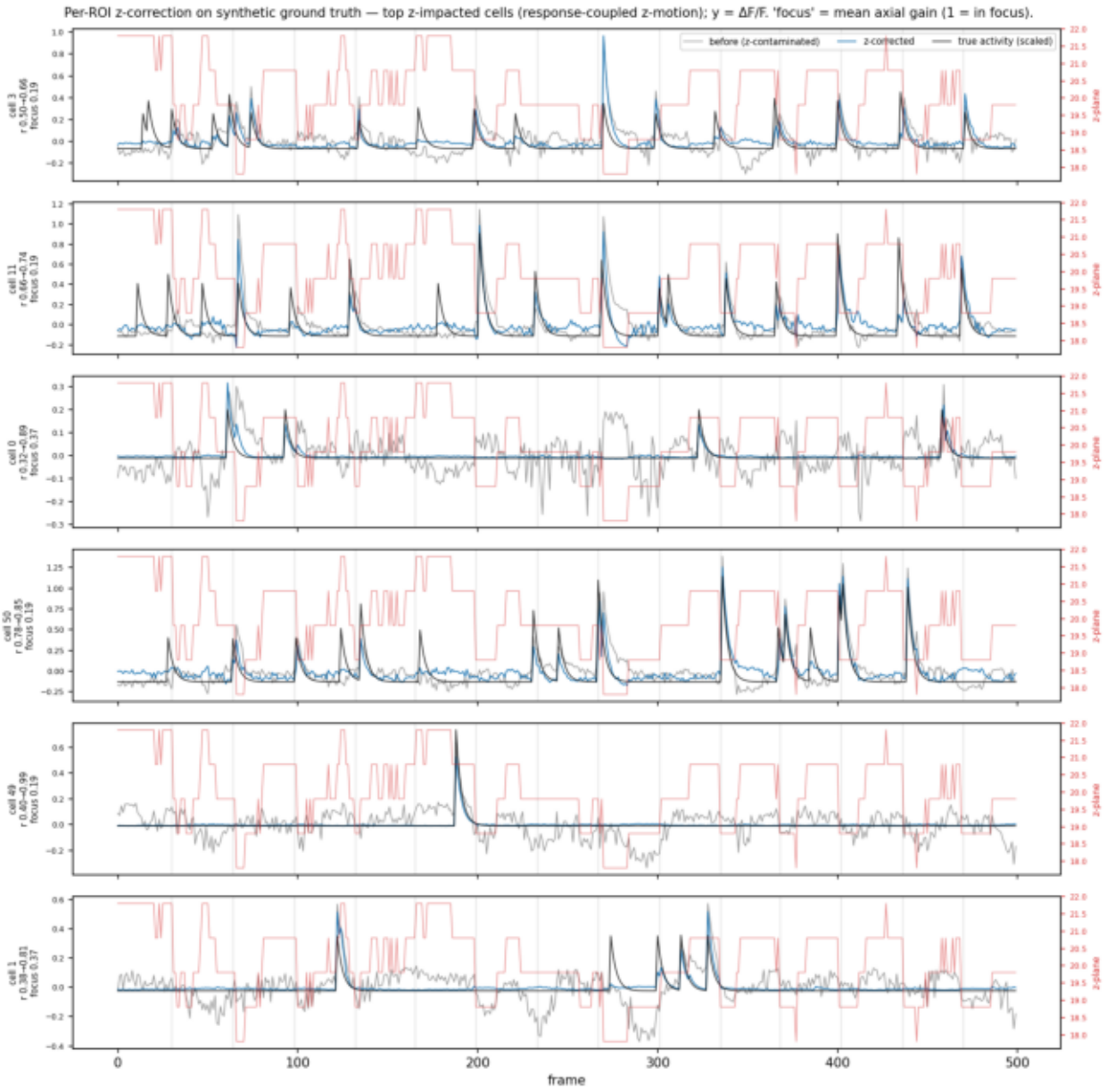
6. Correction — recovery on synthetic ground truth

Known activity is injected into the real z-stack so recovery can be quantified. The z-motion is a staircase (like a real z-position estimate) with downward steps clustered on stimulus responses (grey vertical lines) — the artifact partly overlaps the responses (the hardest case). y = ΔF/F (real amplitudes, not normalised); 'focus' = mean axial gain (1 = perfectly in focus). Rows are the most z-impacted cells among visible ones (focus ≥ 0.15).

Two effects explain the traces — and answer why the uncorrected trace and the correction do not simply track the size of the z-motion:

1. Contamination amplitude is set by the local anatomical brightness-vs-z gradient, not by the step size: when the plane steps to where a bright out-of-focus neighbour overlaps the ROI, the baseline jumps, so even a 1-plane step can give a large ΔF/F swing (grey). F_anat captures this exactly, so the per-ROI subtraction cleans it (grey→blue) and the true transients (black) emerge — the correction can be excellent even when the motion looks small.

2. Recovery is capped by focus: an off-focus cell's true transient is defocus-attenuated in the movie, and subtraction cannot restore signal the objective never collected — so lower-focus cells recover less, and response epochs where the downward step coincides with the transient are the hardest (the artifact and the signal overlap in time).



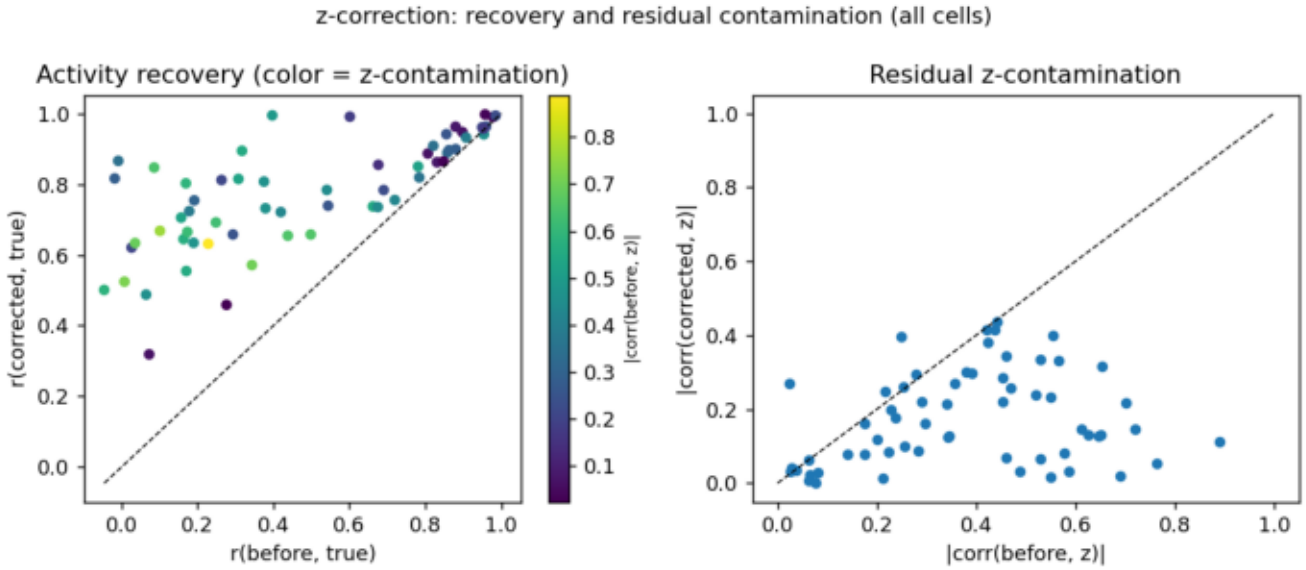
6. Correction on synthetic ground truth — recovery

All cells: recovery r 0.49 → 0.78 | residual z-contamination 0.38 → 0.18

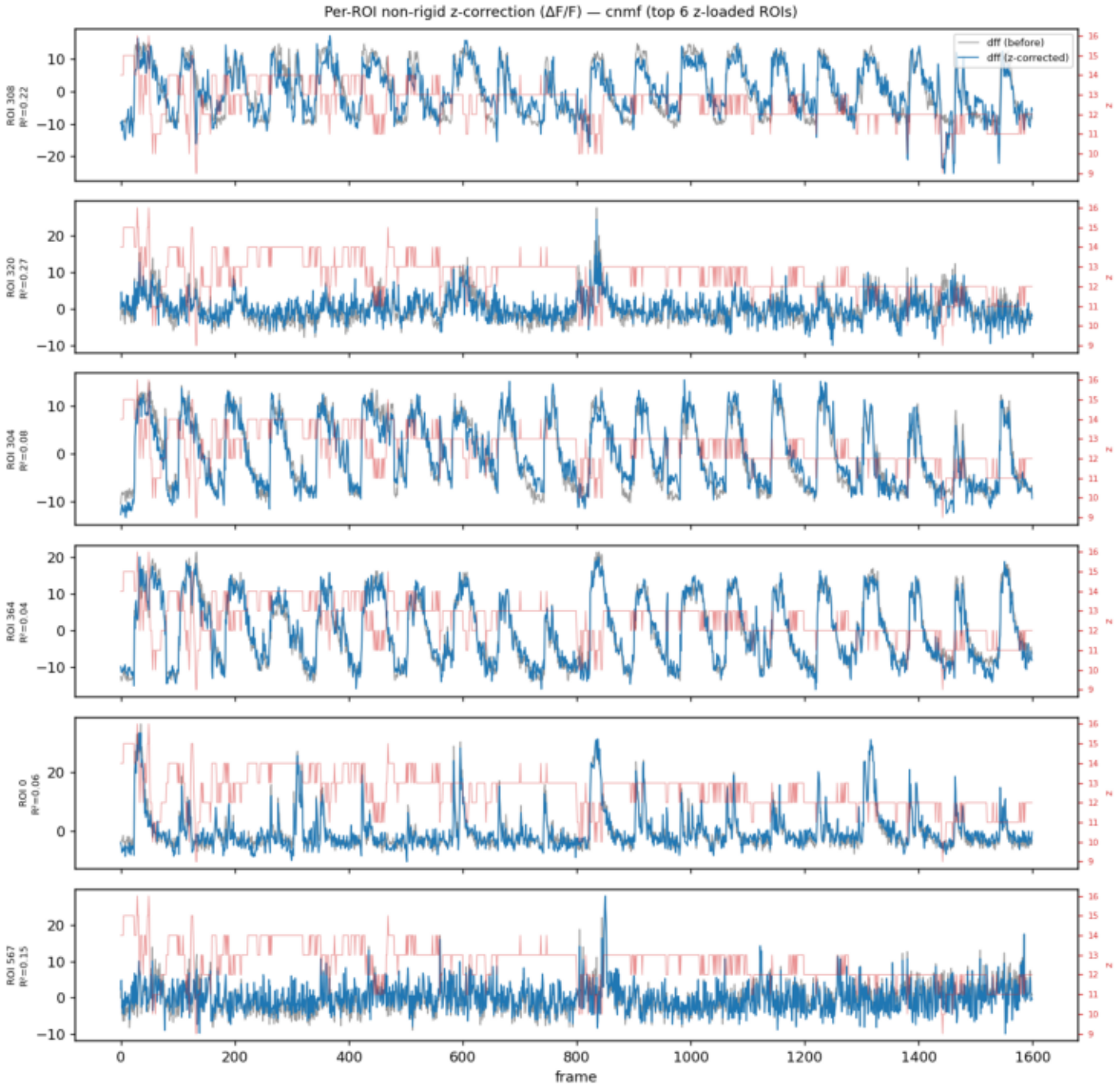
Top z-impacted (10): recovery r 0.63 → 0.83 | residual z 0.49 → 0.23

Benefit: the most z-impacted (visible) cells gain substantial recovery and roughly halve their z-contamination. Well-focused clean cells are left essentially unchanged.

Limit: recovery plateaus below 1.0 — z-defocus attenuates the true signal, which additive subtraction cannot restore, and response-locked motion overlaps the signal.

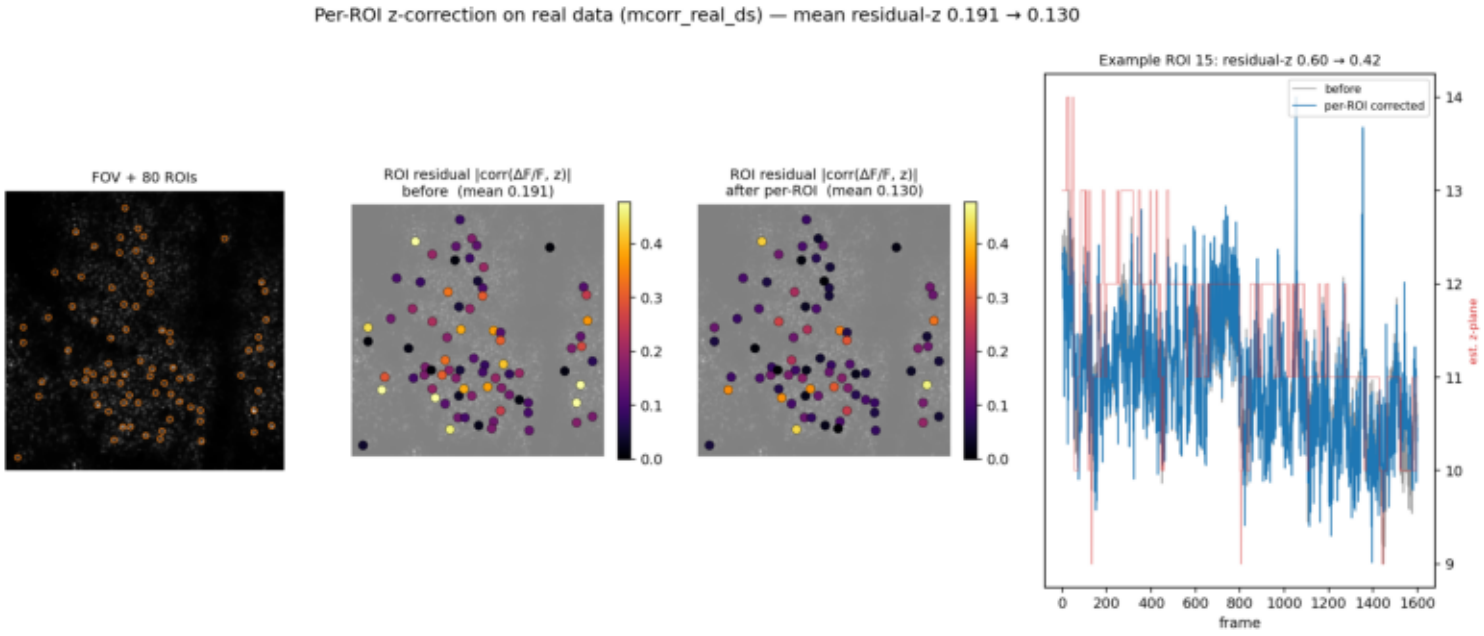


7. Per-ROI correction on real data (CNMF traces)



8. Per-ROI correction on real data (spatial)

The shipped per-ROI correction on a real motion-corrected recording (no ground truth). Left: FOV with extracted ROIs. Middle: per-ROI residual z-contamination |corr(ΔF/F, z)| before vs after correction (brighter = more z-contaminated). Right: an example ROI trace (raw vs corrected) with the estimated z overlaid — the correction removes z-locked fluctuations from the trace. Mean ROI residual-z 0.19 → 0.13 across 80 ROIs.



9. Conclusions & recommendations

What works, and what to do next

Pipeline:

• The z-motion chain is validated end-to-end on real data, both CNMF and Suite2p backends. 7 latent bugs were fixed in the per-ROI path.

z-position estimation:

• Synthetic ground truth: phase correlation (Suite2p) is most accurate (RMSE ~0.7 vs ~1.7 planes for Pearson).

• BUT on real NoRMCorre-corrected movies this does not transfer: phasecorr is much noisier — 94 vs 18 spurious ≥2-plane jumps on the 3-session concat (5×), and a deep-plane bias in single sessions — and edge-cropping collapses Pearson to plane 0. Since z-pos runs after mcorr, phasecorr is not usable there.

• The robust improvement is Pearson + temporal median smoothing of zpos: same drift as legacy but 44 vs 290 spurious ≥2-plane jumps on the real single run.

• Recommendation: keep Pearson (now the default in compute_zcorrel) with temporal smoothing; do NOT adopt phasecorr/edge-crop for production despite their synthetic accuracy. Revisit phasecorr only for raw/lightly-registered data.

F_anat & correction:

• Per-ROI fit-and-subtract removes the most z-contamination at every drift amplitude; per-pixel/per-patch trail. Recovery is capped by defocus attenuation.

• Sub-plane interpolation of the z-peak gives negligible gain (global or F_anat) — not worth adding.

Annex A1. compute_zcorrel — temporal smoothing (legacy vs optimized)

compute_zcorrel_legacy = full-frame Pearson (kept for comparison).

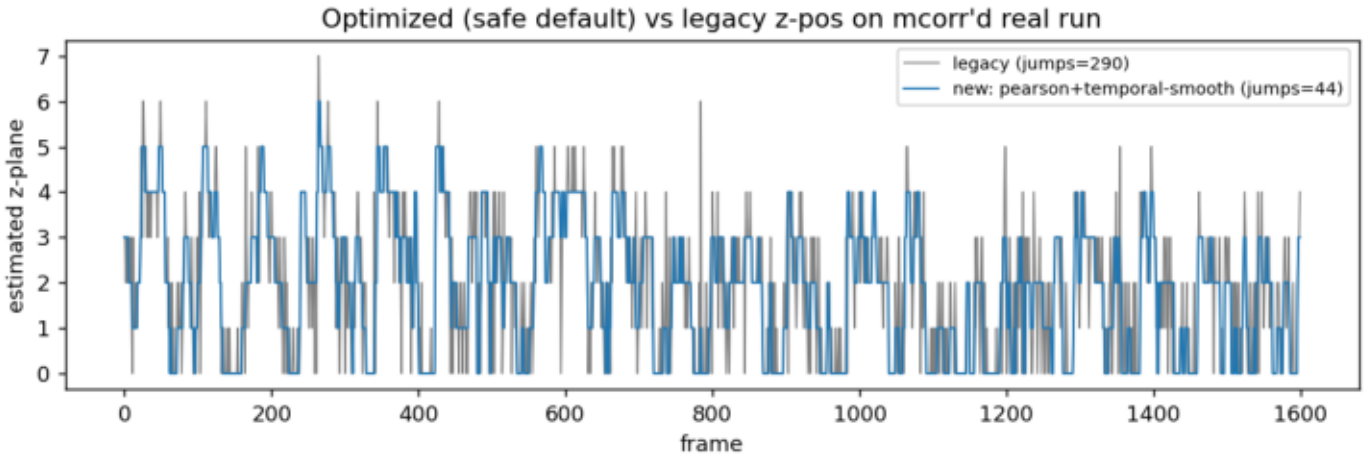
New compute_zcorrel default = Pearson + temporal median smoothing of zpos.

Synthetic ground truth (4 seeds): suite2p/phasecorr is most accurate (RMSE ~0.7 planes) but — see below — fails on mcorr'd data. Pearson+smoothing edges out legacy (r 0.935 vs 0.920).

Two synthetic-motivated options were tested and left OFF by default because they DEGRADE on NoRMCorre-corrected production movies:

- phasecorr (Suite2p): latches onto the deepest low-contrast plane (→ plane 40).
- edge_crop: collapses the Pearson estimate to plane 0.

The one safe win — temporal smoothing — cuts spurious ≥2-plane jumps on the real mcorr'd run from 290 (legacy) to 44, tracking the same drift (r=0.85):



Annex A2. Sub-plane interpolation (tested — negligible gain)

case	metric	integer argmax	sub-plane
global z, amp 1	RMSE planes	2.258	2.261
global z, amp 2	RMSE planes	2.790	2.774
global z, amp 4	RMSE planes	1.808	1.791
F_anat corr, amp 3	recovery r	0.721	0.722
F_anat corr, amp 6	recovery r	0.548	0.549

Parabolic interpolation of the correlation peak was expected to beat integer argmax for both the global z-position and the per-patch depth used to build F_anat. Tested: negligible improvement in both. The error is dominated by discrete multi-plane misassignments and defocus signal attenuation, not ±0.5-plane quantization. Conclusion: sub-plane is not worth adding; improve plane-selection robustness or use the more accurate Suite2p estimator instead.